

SENG 474 — FINAL PROJECT • SPRING 2026

CineEmbed

A multi-modal unsupervised
film recommendation system

All three pre-registered hypotheses PASS • +205 % vs non-deep baseline • Two methodological findings

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Why this problem?

Modern catalogues host 100k+ films. No human can browse them.

- ▶ **Recommendation is the discovery interface** — content-based, collaborative, or hybrid.
- ▶ **TMDb has rich metadata but no user labels** — rules out collaborative filtering at source.
- ▶ **Content-based pipelines are cold-start-robust** — immediate value on new releases.
- ▶ **CineEmbed asks: can purely unsupervised representation learning** produce competitive retrieval on heterogeneous film metadata?

FILMS IN THE CATALOGUE

329,044

564-DIMENSIONAL FEATURE MATRIX

7 modality blocks

numerical · genre · language · decade ·
awards · text · director

z = 32 · L2-normalised cosine · <5 ms query

42 MB index · 107 passing tests

Why unsupervised learning?

No interaction logs. No ratings. Learn structure from item metadata alone.

No user labels available

TMDb dump has metadata only — no clicks, watches, ratings.

0 user IDs in the dump

Eliminates supervised and collaborative methods at the source.

Modality variance imbalance

Ratio 0.046 between text (384-d, dense) and sparse one-hot blocks.

$\text{Var}(\text{text}) / \text{Var}(\text{decade}) \approx 22$

The central data problem — drives every architecture choice.

Test-set hygiene

Three label axes (genre, decade, lang) used only for evaluation, never training.

Single seed = 42 across pipeline

Honest measurement of what the latent has learned.

Dataset • 329,044 TMDb films, 7 modality blocks

#	Block	Input dim	Encoding	Projected dim
1	numerical	6	log-scaled popularity, vote_count, runtime, ratings	16
2	genre	22	multi-label one-hot over 21 TMDb genres + has_genre	16
3	language	31	one-hot top-30 original_language + lang_other	16
4	decade	2	normalised (year – 1900)/130 + has_release_date	4
5	awards	6	log1p of prior Oscar / Palme nominations & wins	16
6	text	384	MiniLM-L12-v2 overview embedding (L2-normalised)	64
7	director	113	PCA-64 of bio + has_bio + 31 lang + 16 country + flag	32
	TOTAL	564		164

MD5 (CANONICAL)

e99cee84...

SEED

42

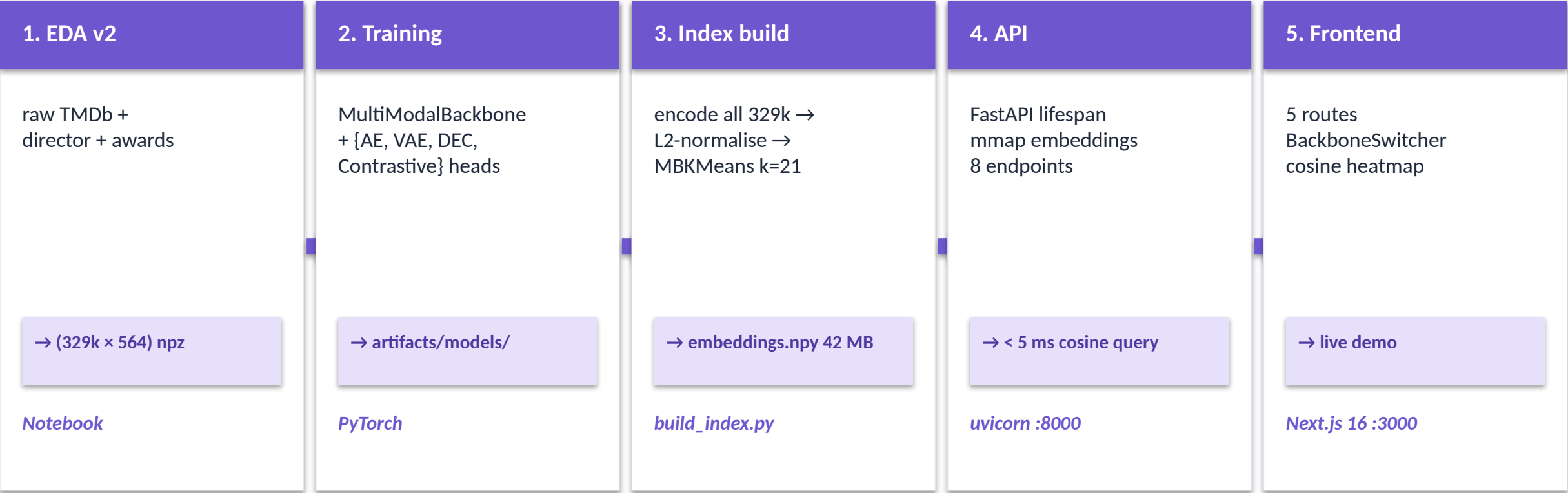
TEXT ENCODER

MiniLM-L12-v2

BIO COVERAGE

3.18 % rows

End-to-end pipeline



Every step is deterministic at seed = 42 · Full reproducibility via 107-test pytest suite (1535 LOC) · All artifacts under artifacts/

FILMS	FEATURES	BACKBONES	CLUSTERS	TESTS
329,044	564 → 32	3 trained	k = 21	107 pass

Multi-modal backbone — per-block projection

Each block gets its own linear projection

numerical[6]	→ Linear(6→16)	→ ReLU → Dropout 0.10
genre[22]	→ Linear(22→16)	→ ReLU → Dropout 0.10
language[31]	→ Linear(31→16)	→ ReLU → Dropout 0.10
decade[2]	→ Linear(2→4)	→ ReLU → Dropout 0.10
awards[6]	→ Linear(6→16)	→ ReLU → Dropout 0.10
text[384]	→ Linear(384→64)	→ ReLU → Dropout 0.20
director[113]	→ Linear(113→32)	→ ReLU → Dropout 0.20

Concat (164-d) → Linear(164→128) → ReLU → Linear(128→z)

L2-normalised at inference → cosine = dot product

< 500 k parameters (test-pinned) · $z \in \{32, 64, 128\}$

Symmetric decoder mirrors encoder per-block.

ae_z32

DEMO BACKBONE

32-d

42 MB index · best gNMI

ae_z64

MVP carry-over

64-d

84 MB index · highest geo_NMI

ae_z128

over-parameterised

128-d

168 MB index · near-dead dim

Loss functions

W2 — variance-balanced block weights

```
w_b = clip(1 /  $\sum_j \text{Var}(X[:,j])$ , 0.1, 10)
```

Each modality block contributes proportionally — stops 384-d text from drowning out 2-d decade.

G2 — masked director loss

```
loss_bio counts only on has_director_bio = 1
```

Only 3.18 % of films have a bio. Without masking, the other 97 % would teach the model to predict zeros.

DEC — clustering refinement loss

```
L = KL(P||Q) + 0.1 · L_recon
```

Pulls each film toward its cluster centre while keeping reconstruction honest (KL on soft assignments).

InfoNCE — contrastive pretext

```
two views via per-row block mask,  $\tau = 0.1$ 
```

Two corrupted copies of the same film must look alike. Masking out genre breaks the target (see ND-1).

TRAINING PROTOCOL

Adam · weight_decay 1e-5 · grad-clip 1.0 · early-stop patience 10 · seed 42 · 90/10 split · W&B logging

Evaluation — two metric families

Clustering family — structure exploration

- ▶ Three orthogonal axes: primary_genre · decade · lang_top10.
- ▶ Per-axis NMI, ARI, AMI (Strehl & Ghosh 2002; Vinh 2010).
- ▶ Composite geo_NMI = $(\text{gNMI} \cdot \text{dNMI} \cdot \text{INMI})^{1/3}$.
- ▶ Per-axis-k sweep (genre=21, decade=12, lang=11).
- ▶ Multilabel macro-NMI over 22 genre columns.

axis cardinality: genre = 21 · decade = 12 · lang = 11

Retrieval family — deployment selection

- ▶ genre@k: fraction of top-k whose primary_genre matches.
- ▶ 316 popular queries × top-5.
- ▶ Eyeball top-5 on 10 canonical films.
- ▶ Angular spread: random_pair_cos_{mean, std}.
- ▶ Latent health: dim_std per latent dimension.

deploy gate: genre@5 > 0.70 · dim_std_min > 0.05 ·
no cluster collapse

The two-family split is the protocol that exposes the F1 finding (NMI / retrieval — see slide 15).

Headline empirical result

BEST DEEP vs BEST NON-DEEP BASELINE

+205%

relative gain on genre NMI

0.332 (dec_z64_k21) vs 0.109 (kmeans_raw_k21)

Pre-registered H2 success criterion: > 10 % over non-deep baseline.

Supporting evidence

+287% genre ARI vs kmeans_raw (0.244 vs 0.063)

+292% lang NMI vs pca_kmeans (0.294 vs 0.075)

+178% lang NMI • multi-modal vs vanilla concat-AE

+99% genre NMI • W2 vs W1 uniform ablation

H1 PASS (DEC > AE +1.2 % NMI) • H2 PASS (this slide) • H3 PASS (0.332 » 0.15 absolute floor)

Three-tier comparison @ $z = 64$, $k = 21$

Tier	Model	genre NMI	genre ARI	decade NMI	decade ARI	lang NMI	lang ARI	val_loss	epochs
Non-deep	kmeans_raw_k21	0.109	0.063	0.233	0.093	0.075	0.026	—	—
Non-deep	pca_kmeans_k21	0.084	0.061	0.224	0.085	0.094	0.042	—	—
Simple	vanilla_ae_z64	0.287	0.247	0.369	0.175	0.095	0.030	0.0126	58
W1 abl.	ae_z64_w1	0.165	0.094	0.367	0.176	0.070	0.026	0.0453	37
Multi-modal	ae_z64	0.328	0.229	0.341	0.211	0.264	0.090	0.0208	69
BEST	dec_z64_k21	0.332	0.244	0.342	0.210	0.294	0.090	0.127 †	21

† DEC val_loss combines KL + recon — not comparable to AE pure-recon.

DEC wins NMI

on 2 of 3 axes (genre, lang)

Vanilla wins decade

0.369 — easy axis, KMeans free

MM wins decade ARI

0.211 — Pareto across architectures

DEC's contribution is compactness (ARI +6.6 %, lang NMI +11.4 %), not new structural information.

Nine hero findings

1. Language wins big

Modality projection lifts lang NMI +178 % vs vanilla concat-AE.

→ [Finding 1](#)

2. W2 is critical

Uniform-weight W1 collapses gNMI 50 %, lang NMI 73 %.

→ [Finding 2](#)

3. Recon ≠ clustering

Lowest val_loss (vanilla) ≠ best gNMI (multi-modal).

→ [Finding 3](#)

4. Decade is easy

All architectures reach decade NMI ≈ 0.34–0.37.

→ [Finding 4](#)

5. Pareto across axes

No architecture wins all six metrics — principled trade-off.

→ [Finding 5](#)

6. Deep > non-deep 3–5×

+205 % vs kmeans_raw_k21 on genre NMI (the headline).

→ [Finding 6](#)

7. DEC sharpens, not discovers

ARI +6.6 % > NMI +1.2 % — compactness gain only.

→ [Finding 7](#)

8. Topology evolves

UMAP: blobs → islands → tight islands (slide 12).

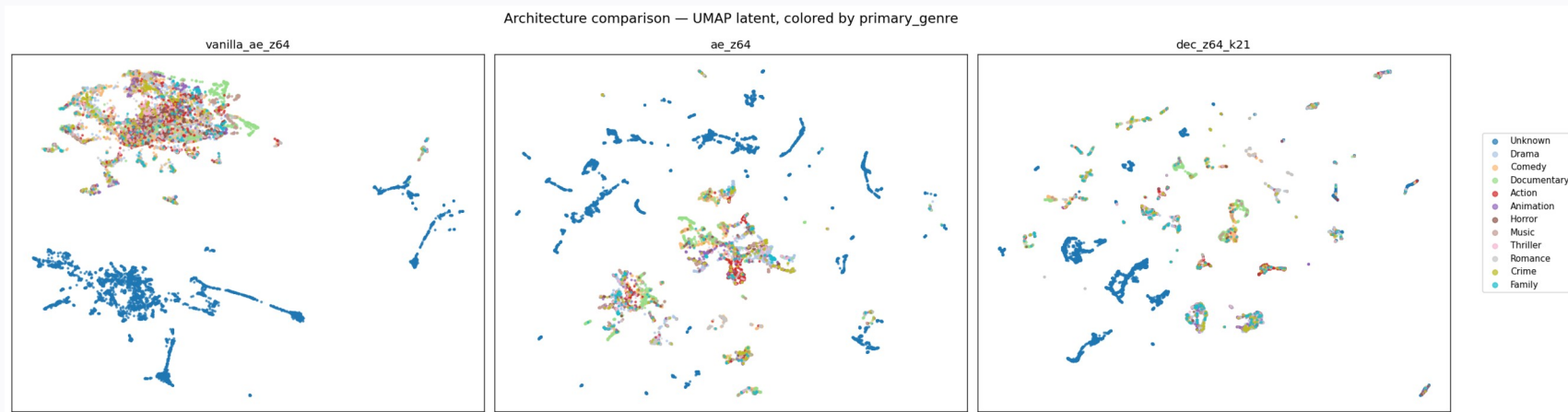
→ [Finding 8](#)

9. Missing-data manifold

Films without release_date form a coherent sub-region.

→ [Finding 9](#)

Latent topology evolves: blobs → islands → tight islands



UMAP • 15k subsample • cosine metric

vanilla concat-AE

2 mega-blobs dominated by Unknown-genre mass

multi-modal AE (W2)

dozens of small genre-coherent islands

DEC (k = 21)

even more atomised, tighter islands

Visual signature of representation learning — the empirical fingerprint of modality-specific projection + KL sharpening.

Ablations — inverse-variance weighting + missing-data

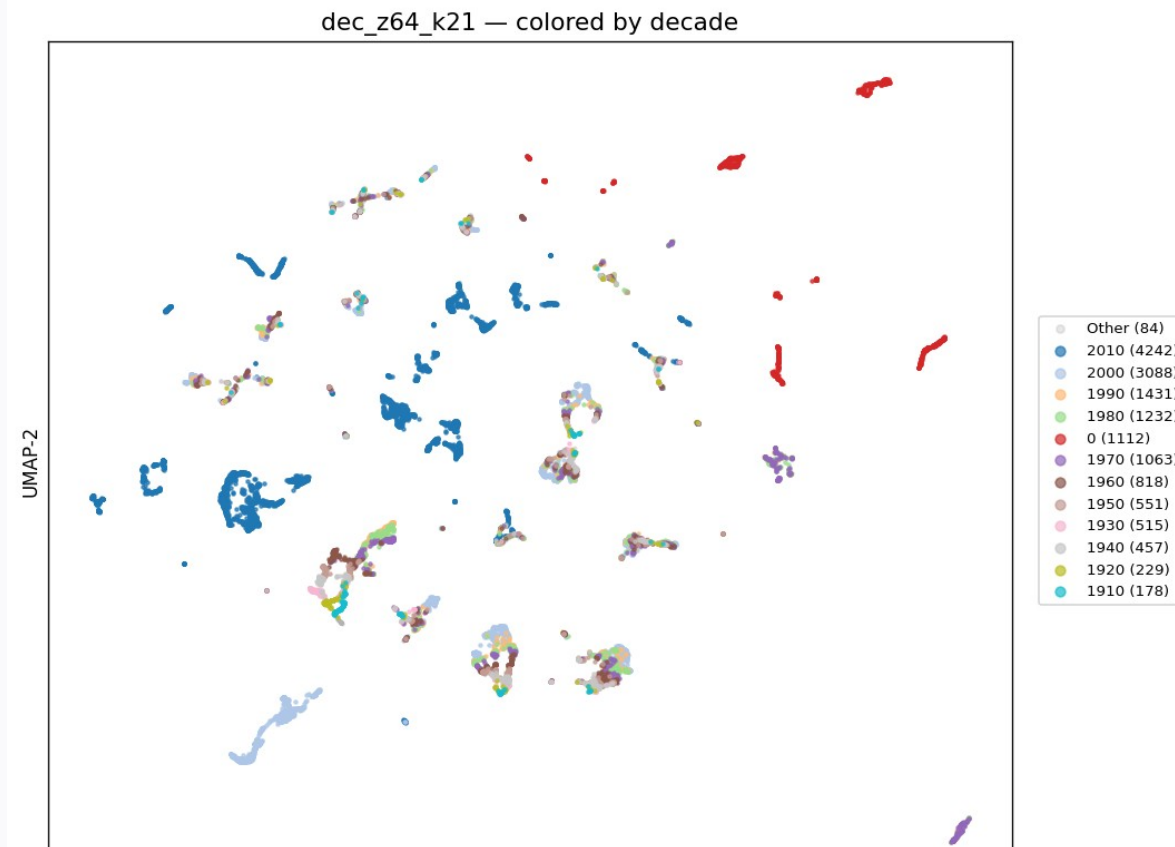
W2 inverse-variance vs W1 uniform

Metric	W2	W1	Δ rel.
genre NMI	0.328	0.165	-50 %
lang NMI	0.264	0.070	-73 %
decade NMI	0.341	0.367	+8 %
epochs early-stop	69	37	—

Asymmetric collapse. Decade (2-d) survives uniform weighting because StandardScaler gave it per-feature var ≈ 1 . Text (384) and language (31) lose all gradient signal.

Model-health diagnostic: W1 patience exhausted at epoch 37.

Finding 9 • missing-release-date manifold



Red = decade_bin 0 (≈ 7.4 % of films, no release date).

Negative results — own them, learn from them

ND-1

Contrastive pretext underperformed

Best Phase 1 gNMI = 0.216 vs MVP ae_z64 = 0.328 (-34 %).

Root cause: modality-dropout includes genre — InfoNCE trains for genre-invariance, the opposite of the clustering target.

ND-2

AE→DEC fine-tune inherited the poison

geo_NMI $\leq$ 0.181 vs MVP DEC = 0.323 (-44 to -67 %).

DEC - KMeans-on-AE-latent delta $\leq$ 0.001 — DEC was fine; the contrastive encoder was already broken.

ND-3

VAE z = 64 likely posterior collapse

Early-stop epoch 12, geo_NMI = 0.127 — below every baseline.

Per-epoch recon vs KL split was not logged — hypothesis strong but not falsified. Top future-work priority.

LESSONS • augmentations must preserve the target • always log recon vs KL split • re-audit selection metric when deliverable shifts

F1 — NMI does NOT predict cosine retrieval quality

DEC TOP-5 COSINES • INCEPTION

≈ 1.000

0.99992 0.99989 0.99986 0.99986 0.99986

difference < 1e-4 (float32 floor)

DEC has the HIGHEST NMI (0.332)

and the WORST retrieval

genre@5 = 0.557 (vs 0.723 for ae_z32)

Mechanism — angular collapse

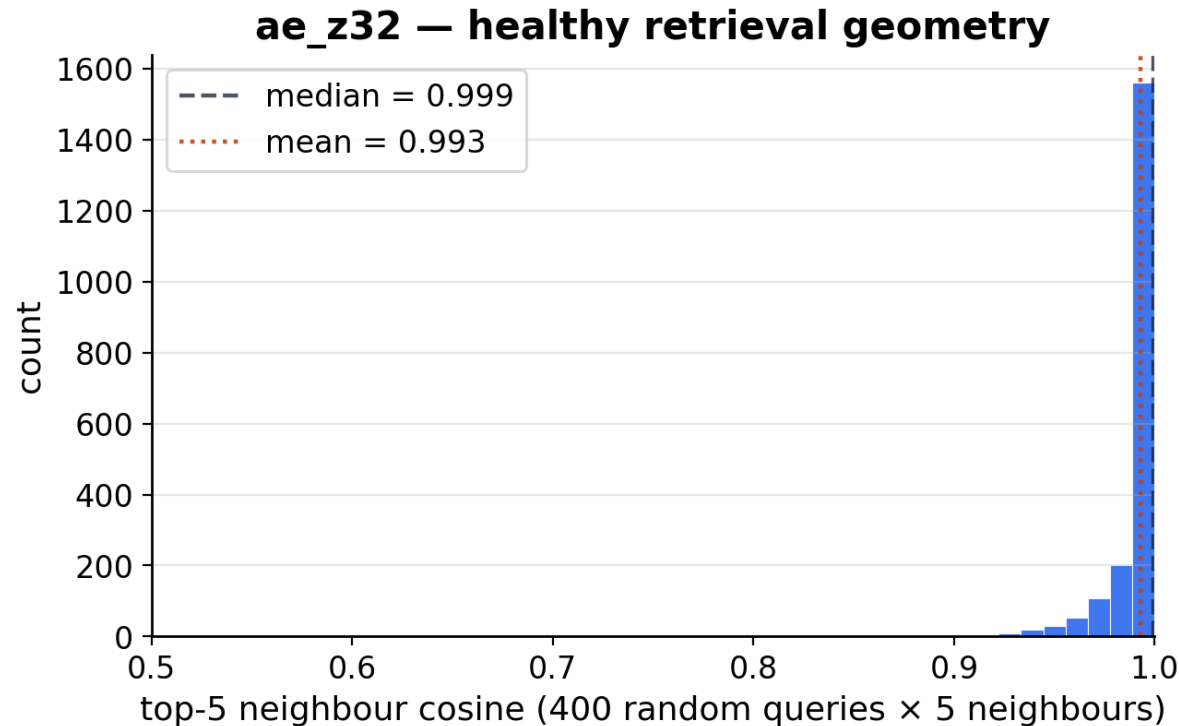
DEC's clustering loss pulls every film inside a cluster toward the cluster centre. After enough training, neighbours sit on top of each other — cosine = 1. NMI only sees cluster IDs (still good); retrieval needs spread inside the cluster (collapsed).

Backbone	gNMI	genre@5	verdict
ae_z32	0.334	0.723	deploy
ae_z64	0.328	0.715	MVP
ae_z128	0.273	0.722	wasteful
dec z64 k21	0.332	0.557	BROKEN

When the deliverable changes, RE-AUDIT the selection metric.

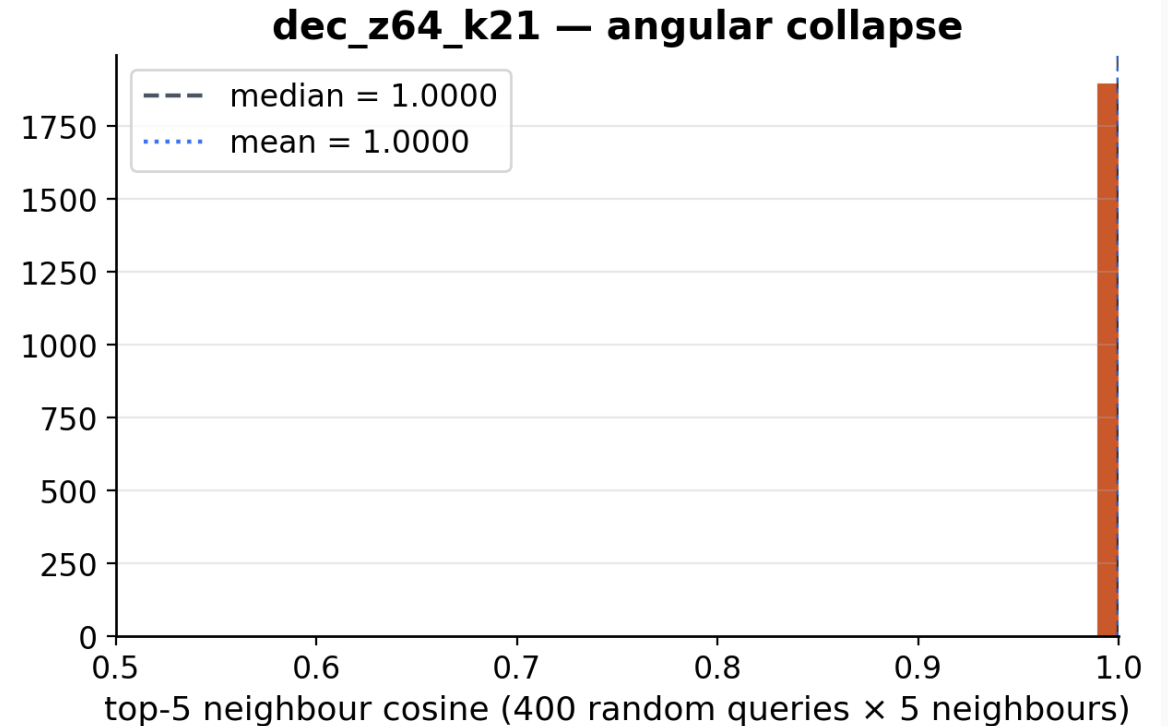
F1 evidence — top-5 neighbour cosines (400 queries)

NMI $\neq$ retrieval — DEC's top-5 cosines saturate ≈ 1.000



ae_z32 — healthy geometry

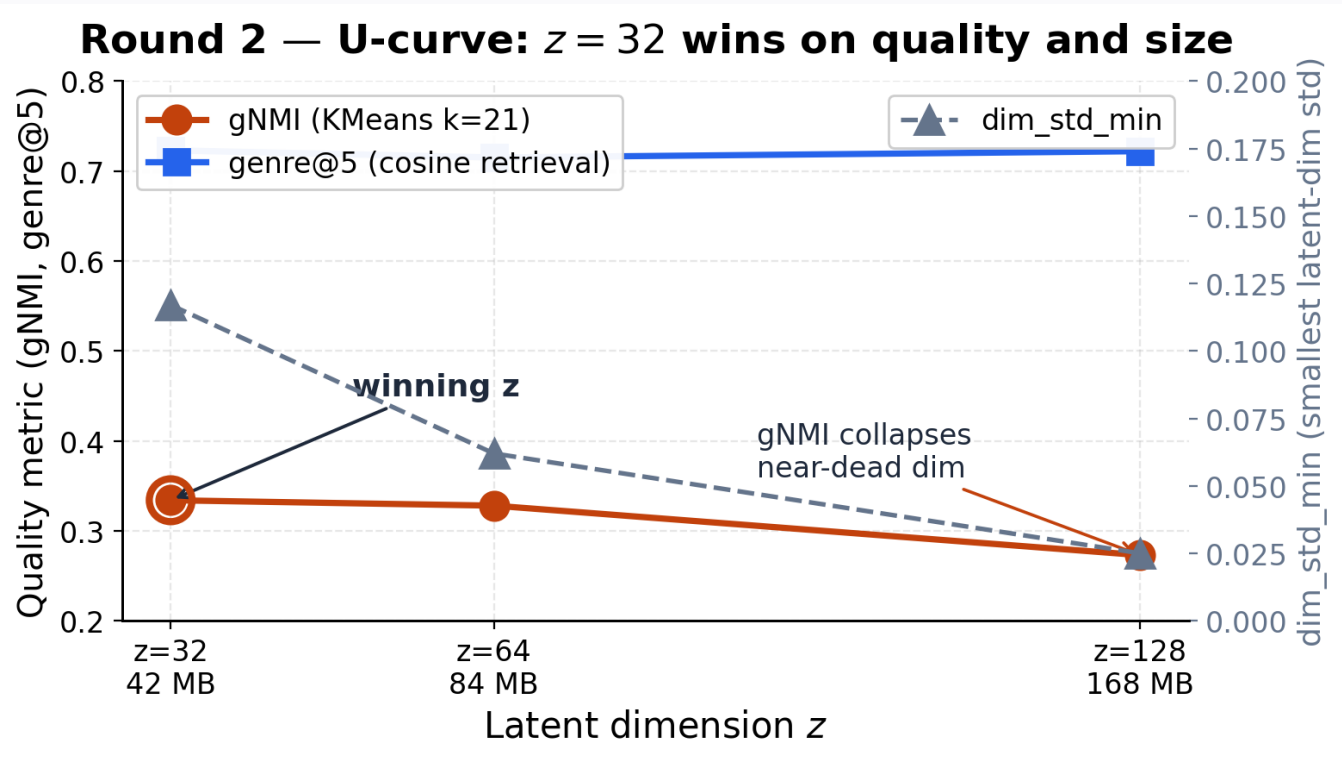
mean 0.993 · median 0.999 · min 0.906 → broad spread



dec_z64_k21 — angular collapse

mean 1.000 · median 1.000 · min 0.998 → saturated

F2 — z = 32 is the right capacity (U-curve)



Smaller and bigger both lose — z = 32 wins

	gNMI	genre@5	dim_std_min
z = 32	0.334	0.723	0.117
z = 64	0.328	0.715	0.062
z = 128	0.273	0.722	0.025

Five signals favour z = 32

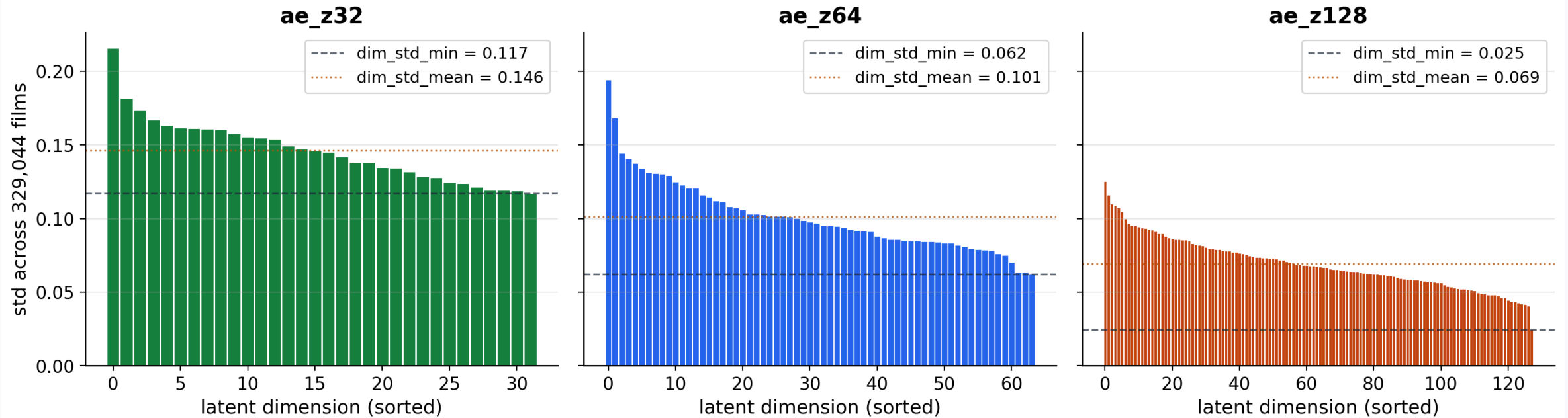
gNMI peak · dim_std_min healthy
pair_cos_std broad · val_loss lowest
Occam — smallest, fastest, 42 MB index

Tight latent forces the encoder to spend its capacity on the information-rich modalities (text, director) and compress the rest.

Decision: demo backbone locked at ae_z32.

F2 evidence — $z = 128$ has near-dead latent dimensions

Per-dimension activation spread — $z=128$ has near-dead dimensions



$z = 32$

healthy

`dim_std_min = 0.117`

$z = 64$

marginal

`dim_std_min = 0.062`

$z = 128$

near-dead

`dim_std_min = 0.025`

Round 2 • AE z-sweep (final selection)

z	gNMI	decade NMI	lang NMI	geo NMI	genre@5	dim_std_min	val_loss	epochs
32	0.334	0.295	0.216	0.277	0.723	0.117	0.0204	100
64	0.328	0.341	0.264	0.309	0.715	0.062	0.0208	69
128	0.273	0.275	0.272	0.274	0.722	0.025	0.0237	53

DEMO BACKBONE LOCKED → ae_z32 • gNMI 0.334 • genre@5 0.723 • 42 MB index • < 5 ms cosine query

GNMI Δ

+0.061

vs z = 128

DIM_STD_MIN Δ

+0.092

vs z = 128

INDEX SIZE

÷ 4

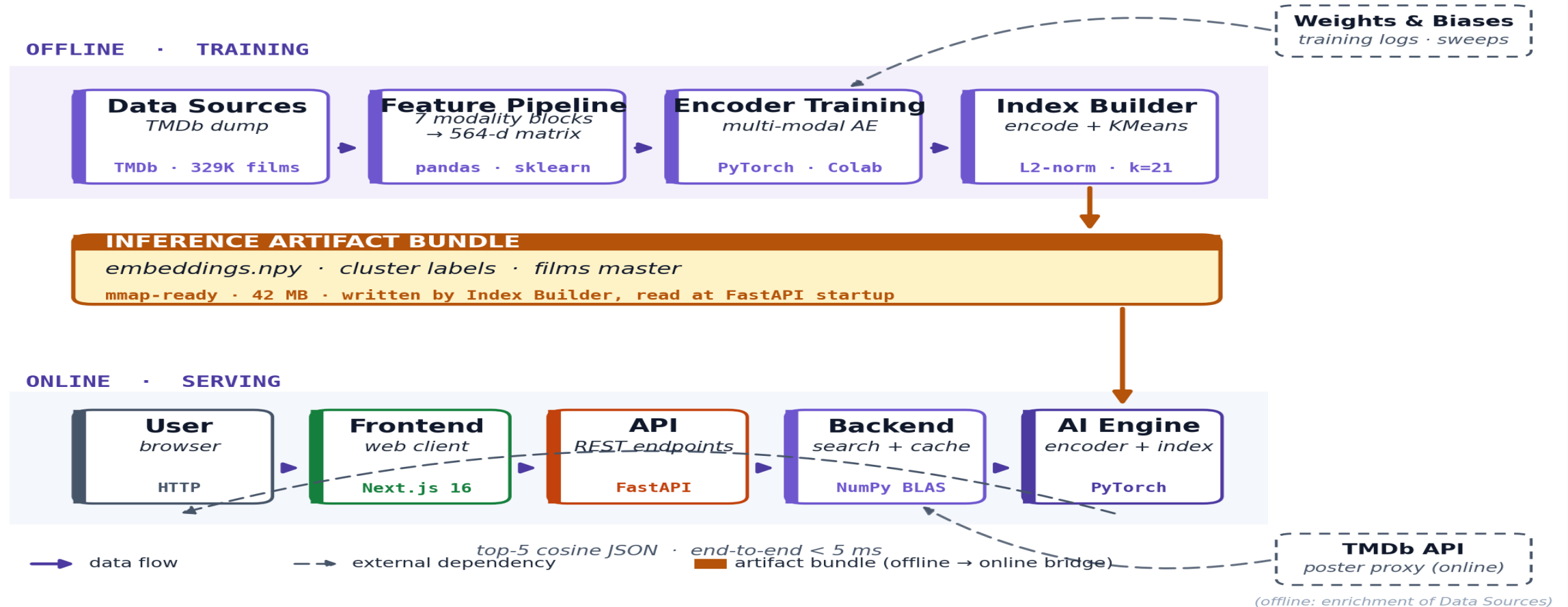
vs z = 128

INFERENCE

× 2

faster

System architecture — how the pieces fit



CATALOGUE

329,044

LATENT

32-d

LATENCY

< 5 ms

TESTS

107

CLUSTERS

21 named

Example recommendations • demo backbone ae_z32

Inception	Spirited Away	Pulp Fiction
1. Interstellar 0.991	1. Princess Mononoke 0.993	1. Kill Bill: Vol. 2 0.987
2. Avengers: Age of Ultron 0.989	2. Nausicaä of the Valley of the Wind 0.991	2. Reservoir Dogs 0.985
3. The Avengers 0.985	3. Pom Poko 0.989	3. Jackie Brown 0.982
4. The Dark Knight Rises 0.984	4. Kiki's Delivery Service 0.987	4. Kill Bill: Vol. 1 0.981
5. Dunkirk 0.983	5. Castle in the Sky 0.985	5. Inglourious Basterds 0.978

Director signatures (Nolan, Miyazaki, Tarantino) recovered without any explicit director label.

Discussion • what worked, what didn't, what's left

What worked

- ▶ Multi-modal projection + W2 weighting — complementary choices.
- ▶ Three-axis NMI / ARI / AMI surfaced the Pareto trade-off.
- ▶ Retrieval sanity check caught the F1 angular collapse pre-deploy.
- ▶ Two-round strategy — 80 % compute saving vs 21-run matrix.
- ▶ Reproducibility: 107 tests • pinned seed • MD5-checked features.

Limitations / future work

- ▶ Modality-dropout failed — target-preserving augmentations next.
- ▶ VAE posterior collapse not falsified — log per-epoch recon vs KL.
- ▶ Single-seed numbers — multi-seed bands would tighten claims.
- ▶ No collaborative signal — two-tower extension natural next.
- ▶ No user study — genre@5 is a content proxy, not user-judged.

References — core literature

Xie, Girshick & Farhadi (2016)

Unsupervised Deep Embedding for Clustering Analysis

ICML — DEC method, target distribution P from Q^2/f_j .

van den Oord, Li & Vinyals (2018)

Representation Learning with Contrastive Predictive Coding

arXiv:1807.03748 — InfoNCE objective.

Vinh, Epps & Bailey (2010)

Information Theoretic Measures for Clusterings Comparison

JMLR — adjusted NMI / AMI definitions.

McInnes, Healy & Melville (2018)

UMAP: Uniform Manifold Approximation and Projection

arXiv:1802.03426 — 2-D topology visualisation.

Wang, Bao, Huang et al. (2020)

MiniLM: Deep Self-Attention Distillation for Pre-Trained Transformers

NeurIPS — sentence-transformers/all-MiniLM-L12-v2 (384-d).

Kingma & Welling (2013)

Auto-Encoding Variational Bayes

ICLR — VAE; ELBO = recon + $KL(q||p)$.

Strehl & Ghosh (2002)

Cluster Ensembles — Knowledge Reuse Framework

JMLR — original NMI definition.

Pedregosa, Varoquaux et al. (2011)

Scikit-learn: Machine Learning in Python

JMLR — MiniBatchKMeans, PCA, clustering metrics.

TOOLS & DATA

TMDb API (themoviedb.org) · PyTorch · Weights & Biases · FastAPI · Next.js · pandoc + xelatex

CONCLUSION

CineEmbed ships a working multi-modal unsupervised film recommender.

All three pre-registered hypotheses PASS (H1/H2/H3).

Two methodological findings beyond the system itself:

F1

NMI $\neq$ retrieval

DEC angular collapse:

top-5 cosine ≈ 1.000

F2

z = 32 wins

Tight latent beats z=64 and z=128

gNMI 0.334 · genre@5 0.723

BEST CONFIGURATION

BACKBONE

ae_z32

gNMI

0.334

genre@5

0.723

INDEX

42 MB

Code: github.com/barandincoguz/CineEmbed

Baran Dinçoğuz · Arda Arvas · Kaan Kaya · SENG 474 — Spring 2026 · TED University